



Generative AI at the Edge

Opportunities and Challenges for the Automotive Industry

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Abstract

The whitepaper acts a comprehensive analysis of the implications and potential applications of deploying Gen AI models (edge/on-device) within the automotive industry. It explores the myriad opportunities that this technology presents, including enhanced customer experiences, efficient resource allocation, sales growth, and accelerated design iterations. It also addresses the data privacy concerns, computation constraints, and the need for robust infrastructure through relevant examples.

As the paper unfolds, it deep dives into the solutions of such challenges, emphasizing the importance of a well-structured approach to develop, deploy, and maintain Large Language Models (LLMs) and Language Vision Models (LVMs) utilizing an LLM-ops framework.

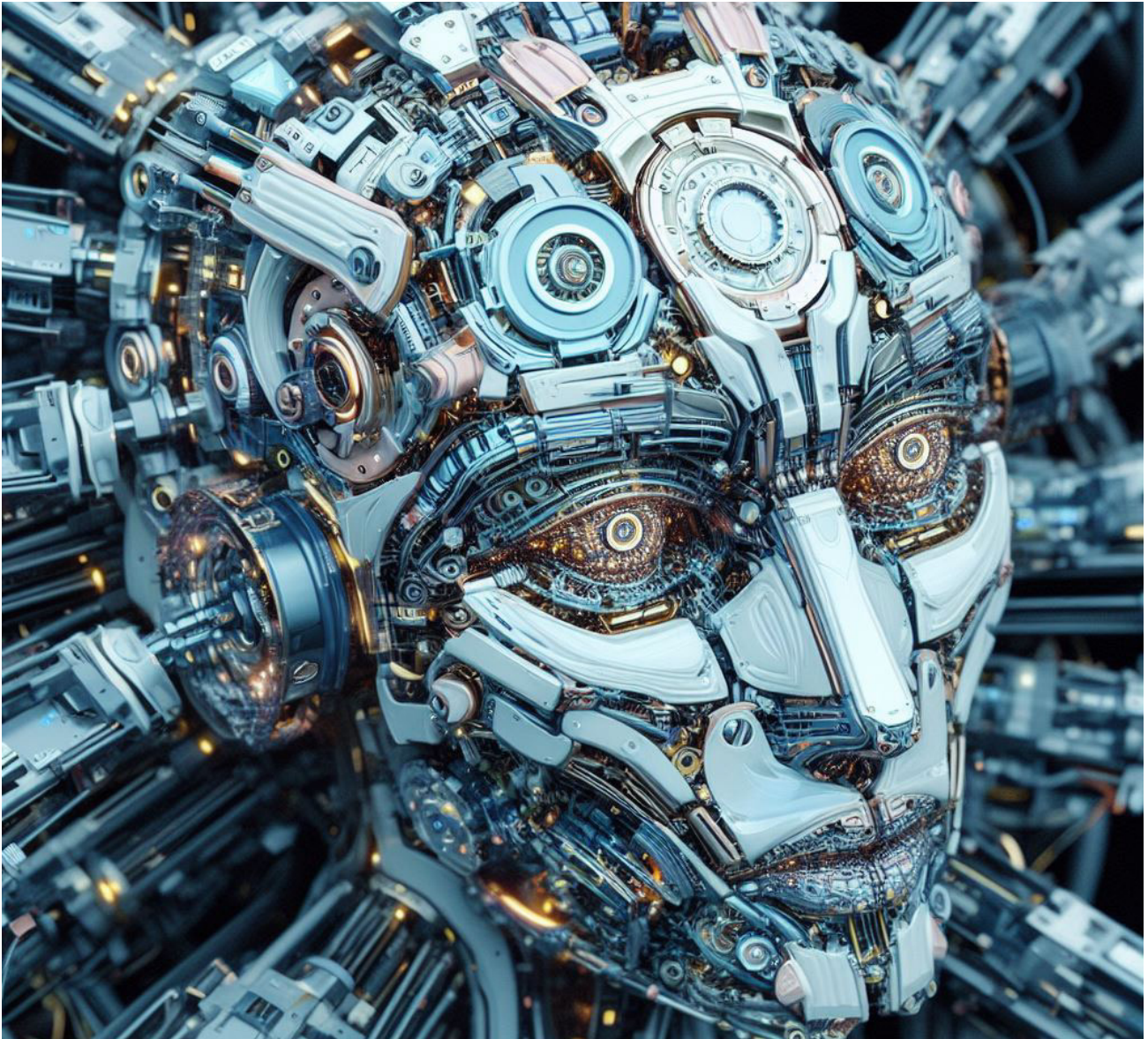
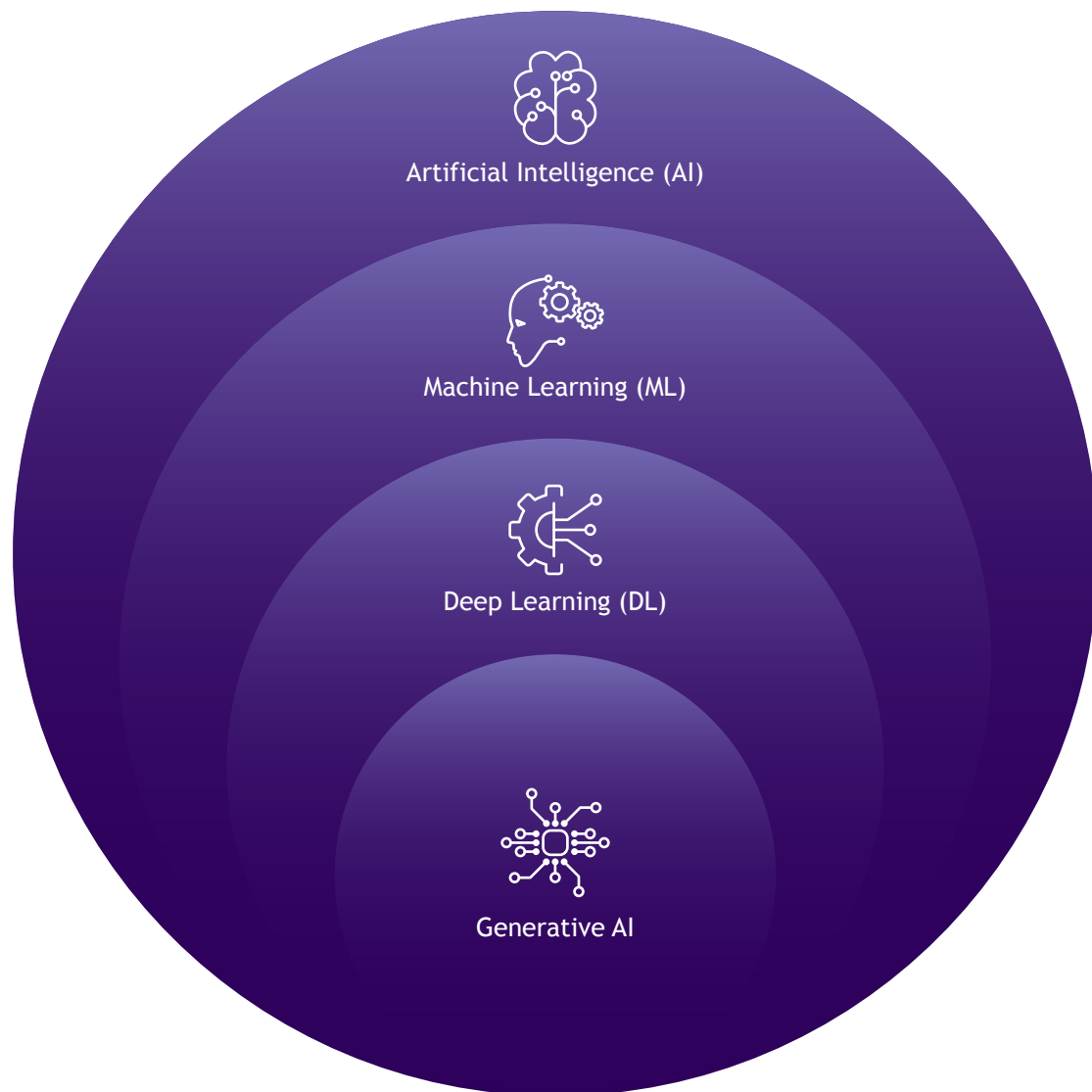


Image generated by DALL-E 3 (a GPT-3 model trained to generate images from text descriptions)

Introduction to Generative AI and Edge Computing

Generative Artificial Intelligence (Gen-AI) has had an unprecedented impact on many industries, revolutionizing processes, and systems since the Generative Pre-trained Transformer (GPT) model was released by OpenAI. In the past, Generative Adversarial Networks (GAN) and Variational Autoencoders (VAE)-based Gen-AI models were used sporadically. However, after OpenAI's ChatGPT took to the market, the use exploded and today, it is used in (almost) everyone's everyday routine.

Generative AI is an Artificial Intelligence that uses Machine Learning systems to create new content, such as text, images, music, audio, and videos. It is powered by foundational models like GPT, which is an implementation of a variant of the transformer architecture, which allows it to generate coherent and contextually relevant sentences. It is trained on a large corpus of text from the internet, and it generates text by predicting the next word or tokens in a sentence.



Before diving into what GPT is and how it works, let's look at its history in general.

Transformers were first proposed in 2016 as a Neural Network Architecture with some unique properties, like:

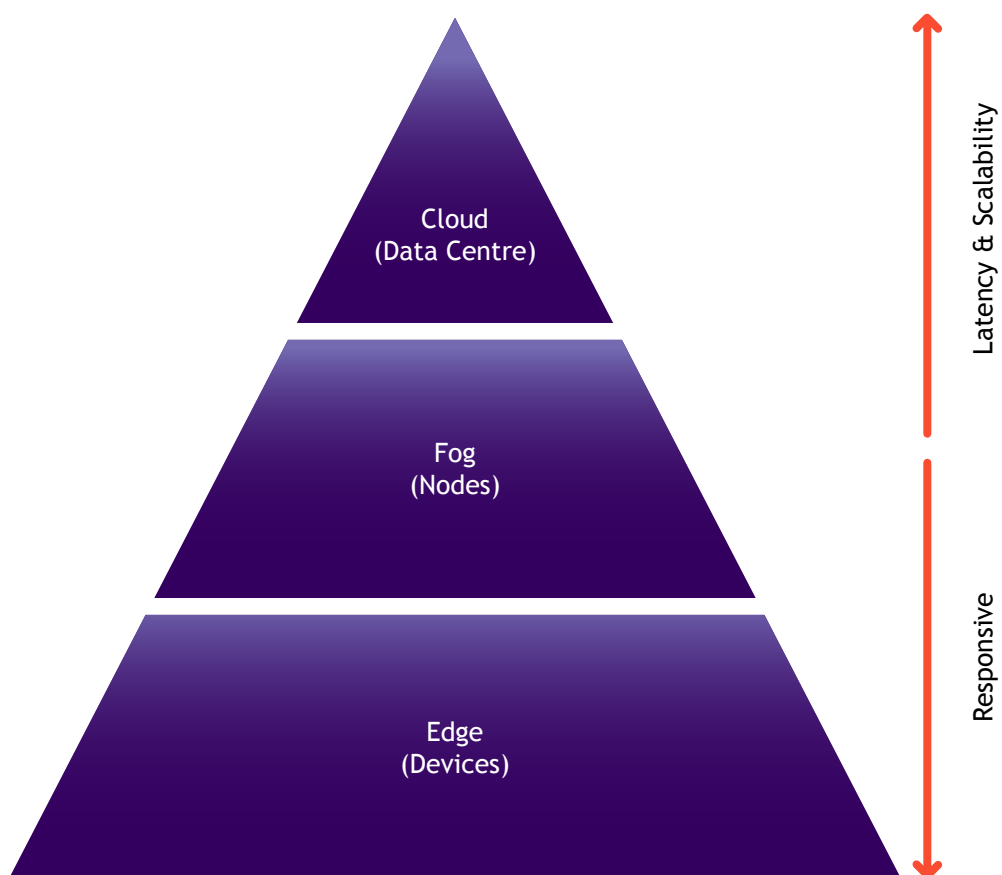
- Generating text that matches the style, tone, and topic of the input
- Ability to reason, evaluate its responses, and take corrective actions.
- It can leverage the parallel computing power of modern hardware, such as GPUs, to speed up the training and inference processes.

The understanding at the time was limited, lacked differentiability, and multimodality until GPT was released in 2016.

This model has been incredibly resilient to various use cases, but it lacks reliability and safety. This has been an inherent problem due to its probabilistic nature, and that is where guard rails are being built by bringing humans into a loop through techniques like reinforcement learning with Human Feedback (RLHF) and sophisticated prompt engineering. The other issues have been with data privacy, ethical and responsible use of technology.

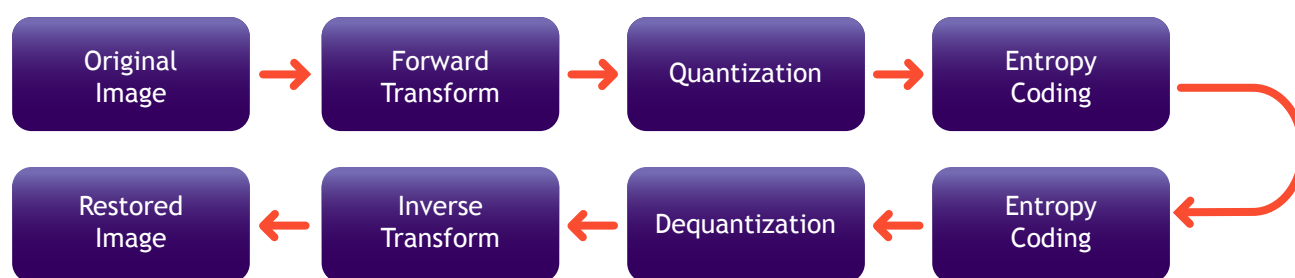
Running Generative AI models on edge devices is picking up pace in recent times. This increase in demand is a result of reduced costs and energy associated with cloud infrastructure. Also, it improves execution latency while increasing data privacy and security. An interesting thing to note here would be that model optimizations are possible in cloud, thus providing scalability for operations.

Edge computing is a distributed computing framework that brings applications closer to data sources such as sensors or actuators. Traditionally, the automotive industry has always run vehicle functions at the edge. Hence, it makes more sense to adopt a hybrid approach where models could be trained or fine-tuned on the private cloud infrastructure and deployed for execution on the edge or at Fog nodes based on the use-case criticality. Fog nodes are devices that provide computing storage and network services between edge devices and the cloud. Fog nodes can be network devices or dedicated servers that coordinate the underlying edge devices.

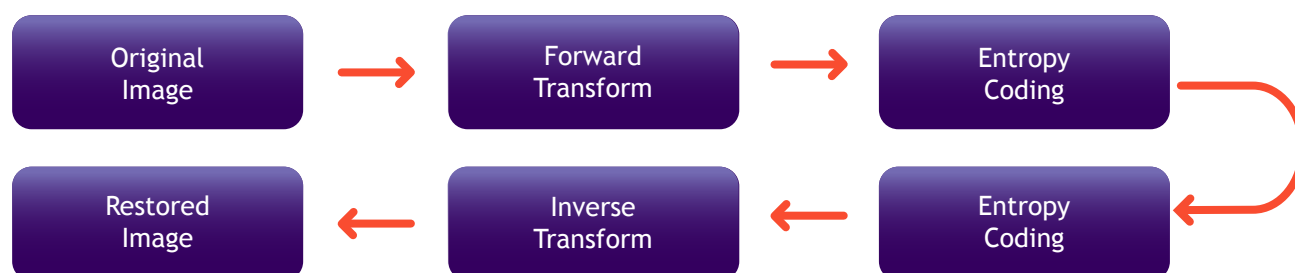


In order to make the large models amenable to running on edge, various techniques have been employed to reduce the size of the model. It is very convenient to use an analogy of lossy compression to a mechanism like quantization, which is used to reduce the model's memory footprint. As shown in the figure, the quantization and lookup functions are used to compress a JPEG, which is a lossy image compression. And decompression of the JPEG file introduces artifacts and errors that distort the original image. Similarly, the GPT model retains much of the information on the web that it has been trained on but provides approximations rather than exact sequences of bits based on the input query it receives. This also enables GPT-based models to be deployed either on the edge.

Lossy Compression



Lossless Compression



The model needs to be continuously monitored for performance, safety, and reliability. This can be achieved by utilizing the LLMOps framework. The table below provides an overview of the current state and future trends of generative AI at the edge of the automotive sector.

DRIVER	CURRENT TRENDS	FUTURE TRENDS	VALUE
Enhancing Customer Experience	Personalization of services, e.g., FF's AI Stack, Mercedes	Reliable prediction of user preferences	Convenience & eco-friendly
National Mandate to Improve Road Safety	Driving support platform, e.g., DriveGPT	Fully Autonomous systems	Saving lives and enhance safety ratings
Faster Product Rollouts	Car Model Design, e.g., Toyota	Developing complete product	Reduce Opex & Capex by 10-20%
Brand Building	Engaging customers meaningfully, e.g., Jasper	Streamlining the entire process chain	Brand Recall Value
Customer Retention	Personal assistant e.g., SoundHound	Multimodal assistance	Improve Customer Satisfaction scores

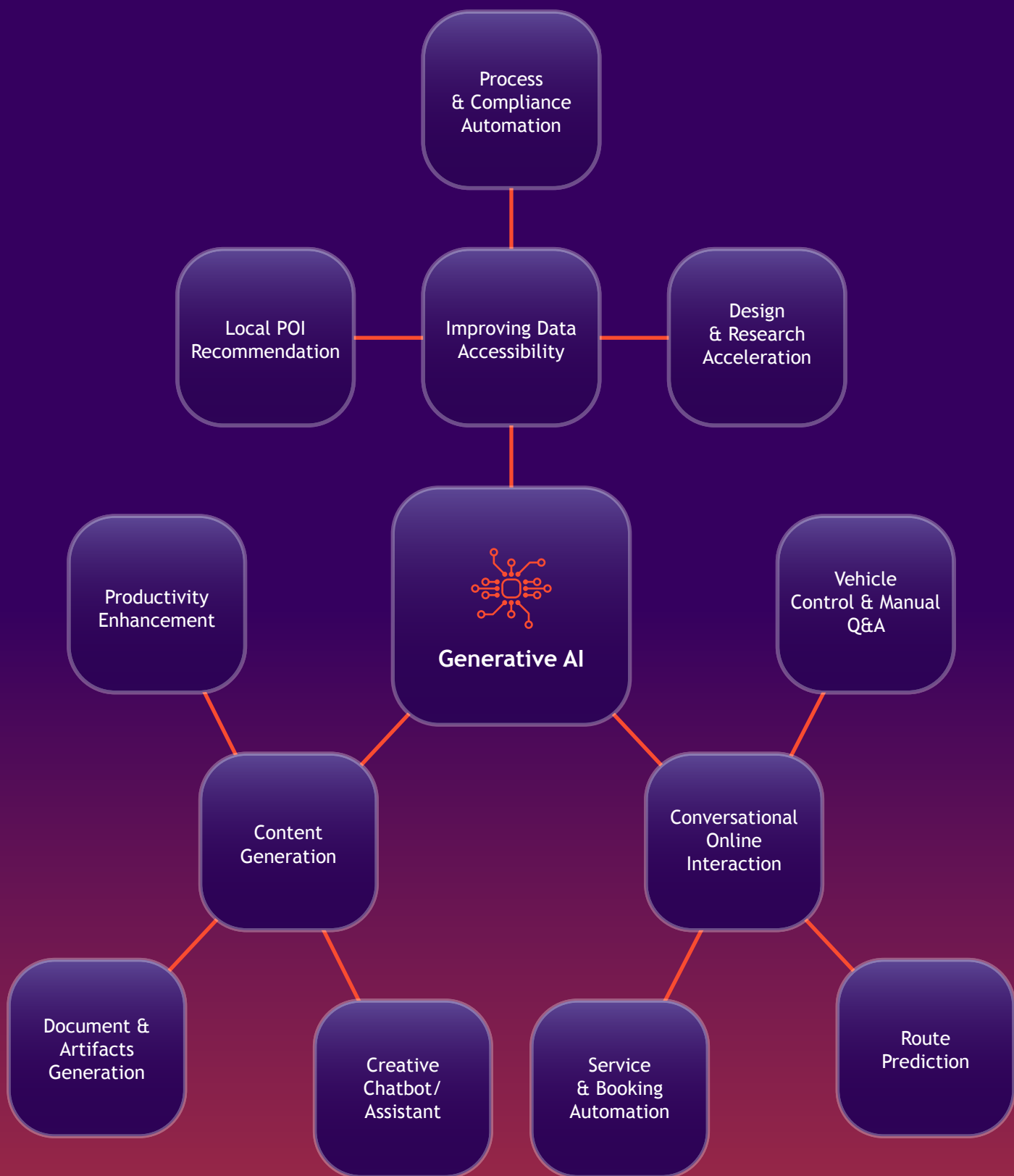
Generative AI Applications for the Automotive Industry

Generative AI is revolutionizing every industry, especially the automotive industry by enabling advanced design, manufacturing, and customer experience enhancements.

This technology offers transformative potential, from streamlining vehicle development processes to creating more personalized and efficient automotive solutions.

Let's look at some of the use-cases and benefits of Gen-AI at the edge of the automotive industry.

- Design and prototyping of new vehicles and components: Gen-AI can be used to optimize the design and manufacturing of vehicles, which can lead to reduced costs. For example, Toyota has been using the Gen-AI technique that allows designers to add initial design sketches and engineering constraints into their creative process.
- Personalization and customization of vehicles and services: Gen-AI creates personalized customer experiences, such as recommending new products and services. For example, Mercedes-Benz has developed voice assistance for its vehicles. Similarly, SoundHound has showcased a conversational, personal assistant.
- Improved vehicle safety: Gen-AI can be used to generate synthetic data that can be used to train safety-critical systems, such as autonomous driving systems and advanced driver assistance systems (ADAS). For example, Tesla has been using Gen-AI in its vehicles to navigate complex driving scenarios by predicting vehicle trajectories.
- Optimizing marketing and branding operations: Gen-AI is being used for creating meaningful customer engagements and tracking marketing spending. For example, Mercedes-Benz is using Gen-AI to offer personalized offers and pricing to customers.
- Other use cases such as local search, safety analysis and predicting vehicle performance can also be enabled using plugins and search techniques with Gen-AI models.



Challenges of Running generative AI models at the edge

There are challenges of generative AI at the edge for the automotive industry from different perspectives, such as:

TECHNICAL

These could range from data availability and quality to model monitoring to hardware and software requirements.

- Compute resources: Gen-AI models can be computationally expensive to run. Therefore, we need to re-spin the processor with additional AI accelerator cores.
- Power: A fully programmable GPU running a Gen-AI model could dissipate 10-watt power per TOPS. This could be even more true for CPUs.
- Data or bus speed: One of the most challenging issues is moving data to and from external memory. TOPS/Watt is also restricted due to low system utilization.
- Interoperability: Gen-AI models cannot be seamlessly deployed from one edge device to another, like any ML/DL model.
- Fine-tuning small vs. large model size: While larger models (with more than 10 billion parameters) produce accurate results, they could be very restrictive to be deployed on a device. A fine-tuned smaller model could outperform a large model, but fine-tuning cost ranging from \$0.0004 to \$0.0300 per 1K token need to be incurred.

BUSINESS

Gen AI for Businesses is all about the value proposition you create for customers, customer satisfaction, competitiveness, user acceptance, and trust.

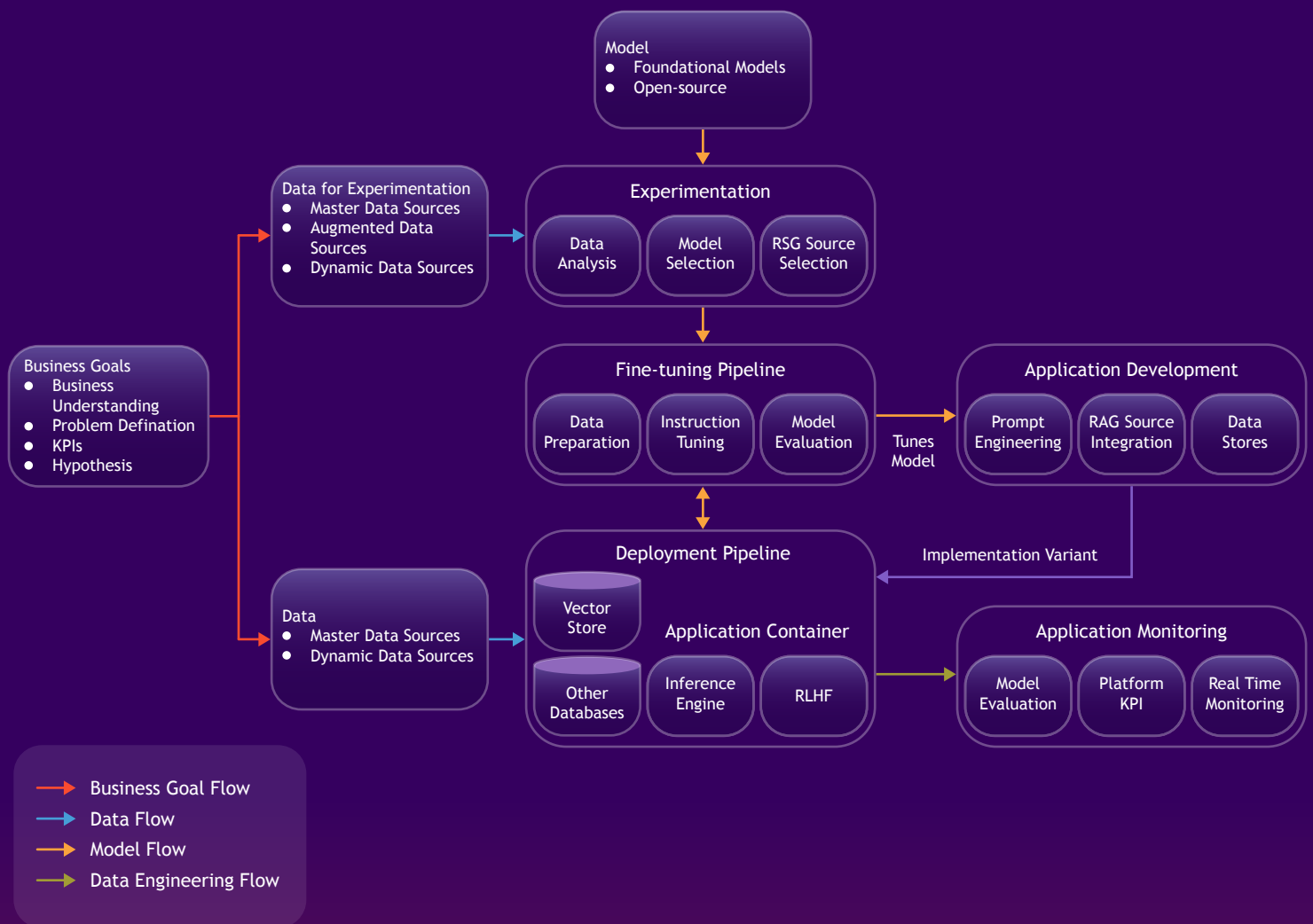
- Communicating value proposition: There is a need to define and communicate the usefulness of Gen-AI-based services to customers. It becomes a real challenge to explain how this could be different or better than the existing alternatives.
- Consistent user experience: While Gen-AI models produce diverse and realistic outputs, they also generate errors, inconsistencies, or low-quality results. Hence, it becomes important for businesses to ensure that their products and services provide a positive and satisfying user experience that is consistent in nature.

SOCIAL

It is interesting to note that Gen AI could be challenging socially considering its environmental impact, ethical implications, social responsibility, security, safety, and compliance.

- Data privacy and security: Gen-AI models can be used to generate data that could be used to track or identify individuals. Also, there is the risk of prompt injection, where bad actors could exploit Gen-AI models to produce harmful content.
- Safety and reliability: Gen-AI models need to be safe and dependable to be used in critical automotive applications. It is difficult to rely on Gen-AI models when they could produce biased or discriminatory outputs called model bias amplification or inadvertently generate incorrect, misleading, or false information, also known as hallucinations.
- Ethical concerns: The use of Gen-AI raises ethical questions about accountability and responsibility for the output generated by these models.
- Regulations: With the increase in accessibility of Gen-AI technology, regulations have become smarter, like the NIST AI RMF, DSA, and EU AI Act, that need to be complied with based on the end-user applications.

Sasken Approach to realize LLMOps for SDV



Conclusion

Currently, the following best practices/frameworks are being used to address the challenges of running Gen-AI models at the edge.

Utilizing LLMOps: This involves Gen-AI/LLM model development, deployment, and maintenance processes. Sasken is deploying Gen-AI models as a container both on the HPC/Edge and the cloud, partnering with a leading Semicon vendor, as can be seen in the above figure.

Responsible AI: This framework is built on five core principles, which are fairness in algorithmic decision-making, transparency in information sharing, accountability for the effects of the system, and ensuring user privacy and safety by complying with regulations.

About the Author

Debashis Panigrahi is the Product Portfolio Manager at Sasken and manages the company's automotive accounts. He works closely with business teams on offerings, solutions, GTM strategy, and competition analysis. He has authored articles on Multimedia, Smart Devices, and Automotive Electronics in several leading trade publications.

About Sasken

At Sasken is at the forefront of Product Engineering and Digital Services, crafting solutions that redefine industries and shape the future of technology. Our deep engineering capabilities and technology patents have transformed Fortune 500 clients and powered billions of devices worldwide. We provide concept-to-market and chip-to-cognition R&D services to global leaders in the Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, Satcom, Telecom, and Transportation Industries.

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